Struct:

Isentropic Problem

{ name :: Symbol = :isentropic

Smooth :: Bool = true

init_gamma

gamma = 3

In this subsection, we will test the accuracy of the Lagrangian type DG scheme (2.9)-(2.10) and the Lagrangian type OFDG scheme (3.6)-(3.8) for the two-dimensional compressible Euler equations. The third order SSP Runge-Kutta time discretization is used. The Courant number λ is set to 0.5, 0.3, and 0.1 respectively when $k = 1, 2$ and 3. For the accuracy tests with $k = 3$, we also replace Δl_{ij} by $(\Delta l_{ij})^{4/3}$ in order to make the temporal and spatial orders match.

Example 4.1 We first consider an isentropic problem with the periodic boundary condition. The initial condition is:

init

$$\rho_0(x, y) = \frac{1 + 0.2\sin(\frac{x+y}{2})}{\sqrt{6}},\ u_0(x, y) = v_0(x, y) = \sqrt{\frac{\gamma}{2}}\rho_0,\ p_0(x, y) = \rho_0^{\gamma},\ (x, y) \in [0, 4\pi]^2.$$

Mesh

init_cons

We set $\gamma = 3$, and we can verify that $\sqrt{6}\rho(x, y, t)$ is the exact solution of the two-dimensional Burgers' equation:

$$\beta_t + \left(\frac{\beta^2}{2}\right)_x + \left(\frac{\beta^2}{2}\right)_y = 0,\quad \beta_0(x, y) = 1 + 0.2\sin\left(\frac{x + y}{2}\right),$$

and the exact solutions (u, v) and p satisfy:

$$u(x, y, t) = v(x, y, t) = \sqrt{\frac{\gamma}{2}}\rho(x, y, t),\ \ p(x, y, t) = \rho(x, y, t)^{\gamma}.$$

The initial computational domain is divided into uniform squares with mesh size $h = \frac{4\pi}{N}$. The final time is $t = 1$.

$\beta_t(x,y) + \beta\beta_x + \beta\beta_y = 0$

$\frac{d}{dt}(\beta(x(t),y(t),t)) = \beta_t + \beta_x x'(t) + \beta_y y'(t) = 0$

if $x'(t) = \beta(x(t),y(t),t)$, $y'(t) = \beta(x(t),y(t),t)$

$x(t) = x_0 + \beta t$

$y(t) = y_0 + \beta t$

$\beta = 1 + 0.2\sin\frac{x_0+y_0}{2} = 1 + 0.2\sin(\frac{x+y}{2} - \beta t) \Rightarrow \beta t(0.8, 1.2)$

Given x, y, t solve β

$f(\beta) = \beta - 1 - 0.2\sin(\frac{x+y}{2} - \beta t)$

$f'(\beta) = 1 + t\,0.2\cos(\frac{x+y}{2} - \beta t)$

$\frac{y-0}{x-x_0} = k$

$y = k(x - x_0)$

$\frac{y}{k} = x - x_0$

$x_0 = x - \frac{y}{k}$

$\beta_{new} = \beta - \frac{f(\beta)}{f'(\beta)}$

$\Rightarrow \sqrt{6}\rho = \beta$, $\rho = \frac{\beta}{\sqrt{6}}$

Use Newton method get exact_cons